

Speed and velocity

Speed is different to velocity. How quick you are going is easy to work out – divide how far you travel by the time it takes. Your velocity is how fast you travel in a particular direction. Changing direction without slowing reduces your velocity, but your speed stays the same.

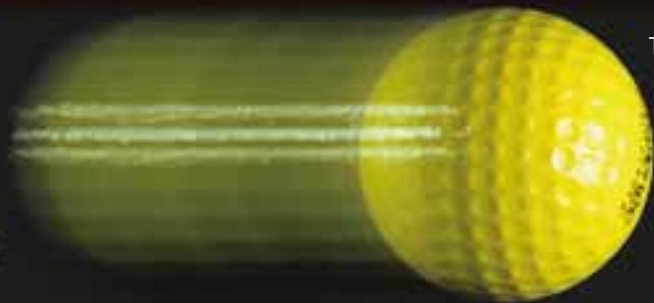
*If you drive 80 km (50 miles) in two hours,
your speed is 40 kph (25 mph).*

Accelerating is fun, but in science it can be confusing. This is because acceleration doesn't just mean speeding up. It is any change in velocity. So, it is also used to describe slowing down and changing direction.



Inertia

When things are standing still or moving, they are quite happy to continue with what they are doing. This stubbornness is called inertia – it is the object's resistance to change.



The golf ball will carry on rolling until friction, gravity, and air resistance slow it down.

Rescue helicopters balance forces so they can hover above the waves.



Become an expert

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Balanced forces

Forces act on objects all the time. Opposing forces can be balanced out. When this happens the object won't be pushed in any direction.